Human IVD scRNA-seq Multi-Dataset Integration Analysis

173,628 cells | 7 datasets | 29 donors | 4 degeneration grades

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1. Overview

This report summarizes an integrated single-cell RNA-seq analysis of human intervertebral disc (IVD) tissue across 7 publicly available GEO datasets, spanning healthy donors and patients with mild, moderate, and severe disc degeneration. The goal was to characterize IVD cell types, identify how cell states and gene expression change with degeneration, and map altered signaling networks.

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2. Datasets

GEO Accession	Study	Tissue	Condition	Samples	Cells (post-QC)
GSE160756	Gan/Liu 2021	NP, AF, CEP	Healthy atlas	7	~60,000
GSE199866	Cherif 2022	NP	Paired degen/non-degen	4	~8,000
GSE244889	Wang 2023	NP	Mild vs severe	7	~45,000
GSE255768	Shi 2024	CEP	Degeneration	2	~5,000
GSE233666	Guo 2023	NP	Immune/ossification	4	~20,000
GSE205535	Li 2022	NP	Normal + degenerated	2	~10,000

GSE189916	Jiang/Sheyn 2022	IVD	Neonatal + adult	6	~25,000

QC: MAD-based filtering per sample (nGenes, nCounts, %mito); Scrublet doublet removal. 173,628 / 222,433 cells retained (78%).

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3. Integration

- **Batch correction:** Harmony on dataset + donor_id (29 donors, 7 datasets)
- **Neighbor graph:** FAISS approximate kNN (k=30, IVFFlat index)
- **Clustering:** Leiden at res=0.3/0.5/0.8/1.2 (9/12/19/27 clusters)
- **Gene space:** 25,304 genes present in ≥3 datasets; 4,000 batch-aware HVGs

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4. Cell Type Annotation

12 clusters identified at Leiden res=0.5, manually annotated using IVD-specific marker panels:

Cell Type	Cluster	Key Markers	n cells	%
NP: canonical	0	ACAN, COL2A1, SOX9, COL9A3, SCRG1	47,120	27.1%
NP: degenerative (UPR)	3	SQSTM1, DNAJB9, TNFRSF12A, EIF1	34,810	20.1%
NP: MT-high	5	MT1G, MT1E, MT1X, MT2A, MALAT1	31,502	18.1%
NP: HAPLN1+	1	HAPLN1, FN1, TIMP3, FGFBP2	17,040	9.8%
AF fibroblast	4	COL1A1, COL1A2, COL3A1, SCX	16,448	9.5%
NP: stress response	2	JUN, FOS, GADD45B, DNAJB1	9,967	5.7%
Pericyte/SMC	7	TAGLN, MYL9, CALD1, NR2F2	4,068	2.3%

Macrophage	10	CD68, CD14, TYROBP, CTSS	3,482	2.0%
Endothelial	8	PECAM1, VWF, CDH5, SPARCL1	3,382	1.9%
Monocyte/Neutrophil	11	LYZ, S100A8, S100A9, MNDA	2,511	1.4%
Erythrocyte	9	HBB, HBA1, HBA2, AHSP	1,658	1.0%
T/NK cell	6	CD3D, CXCR4, GNLY, NKG7	1,640	0.9%

Key figures: `05_annotation/umap_annotated_v2.png`, `05_annotation/dotplot_markers.png`

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5. Compositional Analysis

Cell type proportions computed per donor (n=29), tested with Kruskal-Wallis across 4 conditions.

- **AF fibroblast** is the only cell type with significant compositional change (p=0.038), increasing from ~7% (healthy) to ~24% (severe) — consistent with fibrocartilaginous metaplasia.
- NP state proportions show high inter-donor variability, limiting statistical power with n=29 donors.
- Within the NP lineage: **NP: HAPLN1+** expands +10.1% in severe; **NP: MT-high** contracts −12.5%.

Key figure: `06_composition/composition_overview.png`

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6. Pseudobulk Differential Expression (DESeq2)

Pseudobulk aggregation per donor × cell type; DESeq2 on 6 cell types × 3 contrasts (severe/moderate/mild vs healthy).

Severe vs Healthy — DEG summary

Cell Type	Total DEGs	Up	Down
NP: canonical	2,641	332	2,309
NP: degenerative	2,331	241	2,090
NP: HAPLN1+	1,786	223	1,563
NP: stress	1,081	104	977
AF fibroblast	1,298	103	1,195
NP: MT-high	953	81	872

Dominant pattern: Massive transcriptional downregulation in severe degeneration across all cell types (~7:1 down:up ratio). Key upregulated genes: ADAMTS5, FN1, TNF, CXCL8, IL6. Key downregulated: ACAN, COL2A1, HAPLN1, COMP, CILP.

Key figure: `07_pseudobulk/volcano_severe_vs_healthy.png`
Data: `07_pseudobulk/all_DEGs_severe_vs_healthy.csv`

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7. Pathway Enrichment (GSEA)

GSEA on ranked gene lists (sign(LFC) × −log10(padj)) using MSigDB Hallmarks + Reactome (IVD-relevant subset). 70–85 significant pathways per cell type.

Consistently downregulated in severe degeneration (all/most cell types)

- **Wnt signaling** (TCF-dependent Wnt, Signaling by Wnt) — loss of chondrocyte maintenance
- **Notch signaling** (Signaling by Notch, Pre-Notch processing) — loss of progenitor niche
- **Cellular senescence** (Oxidative stress-induced senescence, SASP, DNA damage senescence)
- **ECM organization** (Extracellular Matrix Organization, Collagen Formation)

- **RUNX1/2 transcription** — loss of chondrogenic transcription factor activity

Upregulated in severe degeneration (selected cell types)

- **TNFα signaling via NF-κB** (NP: canonical, NP: degenerative) — inflammatory activation
- **Inflammatory response** (NP: stress, NP: degenerative)
- **Epithelial-mesenchymal transition** (NP: degenerative) — fibrotic shift
- **Collagen crosslinking** (NP: stress, NP: degenerative, NP: MT-high) — matrix stiffening
- **Glycolysis** (NP: canonical) — metabolic reprogramming under hypoxia

Key figure: `08_pathways/gsea_heatmap_severe_vs_healthy.png`

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8. NP Lineage Trajectory (PAGA)

PAGA connectivity analysis on 140,439 NP lineage cells.

PAGA connectivity (key edges):

- NP: stress ↔ NP: degenerative: **0.76** (strongest connection)
- NP: canonical ↔ NP: HAPLN1+: **0.65**
- NP: stress ↔ NP: MT-high: **0.51**

Interpretation: The NP stress-response state acts as a transition hub between canonical NP chondrocytes and the degenerative UPR state. The HAPLN1+ subtype (matrix-organizing) expands in severe degeneration, possibly representing a compensatory remodeling response.

Note: Diffusion pseudotime (DPT) was weakly resolved in this dataset (all DC variances = 7.12e-06), likely due to the large, well-mixed dataset compressing diffusion kernel eigenvalues. Spearman correlation of pseudotime with degeneration grade was significant but weak ($\rho=0.24$, $p<10^{-300}$). PAGA connectivity is the more reliable trajectory metric here.

Key figure: `09_trajectory/np_trajectory_overview.png`

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9. Cell-Cell Communication (LIANA)

LIANA rank_aggregate (consensus resource) on healthy (n=99,883 cells) vs severe (n=19,545 cells) subsets. 39,530 unique ligand-receptor pairs evaluated.

Top lost interactions in severe degeneration

- **TIMP1 → CD63**: Dominant lost interaction across all cell type pairs. TIMP1 (tissue inhibitor of metalloproteinases) signals through CD63 to suppress MMP activity. Loss of this interaction in severe degeneration is consistent with the known increase in matrix degradation.

Top gained interactions in severe degeneration

- **FN1 → CD44/C5AR1**: Fibronectin signaling to macrophages (via CD44 and complement receptor C5AR1) — consistent with inflammatory macrophage recruitment and activation.
- **FN1 → ITGA6** (NP:HAPLN1+ → Endothelial): Fibronectin-integrin signaling promoting angiogenesis.
- **COL1A2 → CD93** (AF fibroblast → Endothelial): Collagen-endothelial interaction, potentially driving vascular ingrowth.
- **SEMA4A → PLXNB1**: Semaphorin signaling gained in NP cells — role in axonal/vascular ingrowth.

Overall pattern: Global increase in total interaction strength in severe degeneration (all NP states show increased outgoing signaling), driven by FN1, inflammatory cytokines, and matrix-degradation signals. Protective TIMP1 signaling is lost.

Key figure: `10_cellchat/liana_communication_overview.png`

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10. Key Biological Conclusions

1. **Five NP cell states** coexist in the IVD, representing a spectrum from canonical chondrocytes to stress-response, UPR-driven degenerative, and metallothionein-high oxidative stress states.
2. **Degeneration is characterized by transcriptional collapse**: 7:1 down:up DEG ratio in severe vs healthy, with loss of ECM maintenance genes (ACAN, COL2A1, HAPLN1, COMP) and chondrogenic transcription factors.
3. **Four pathways are consistently suppressed across all NP states**: Wnt signaling, Notch signaling, cellular senescence programs, and RUNX transcription — suggesting a coordinated loss of tissue homeostasis machinery.
4. **Inflammatory activation is cell-state specific**: TNF/NF-κB and EMT upregulation is strongest in NP: canonical and NP: degenerative states, not uniformly across all NP cells.
5. **AF fibroblast expansion** is the most robust compositional change (p=0.038), increasing from 7% to 24% in severe degeneration — consistent with fibrocartilaginous replacement of NP tissue.
6. **TIMP1→CD63 loss** is the dominant signaling change in degeneration, implicating reduced MMP inhibition as a key driver of matrix degradation. **FN1→macrophage** signaling gain suggests inflammatory amplification via fibronectin fragments.

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11. Limitations

- **Unbalanced condition groups**: 100,234 healthy vs 19,610 severe cells; 15 healthy donors vs 3 severe donors limits statistical power for compositional and DE analyses.
- **DPT trajectory**: Diffusion pseudotime was weakly resolved; PAGA connectivity is used as the primary trajectory metric.